From setup to first insight.

Windows launcher and project explanation
Prepared 30 October 2026

01 / Run everything with one command

Extract the project ZIP to a normal writable folder. Open PowerShell or the VS Code terminal. For this workspace, run:

```
cd "C:\Users\Alok\Desktop\stuff\bhoomi"
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\run.ps1
```

If you move the project, change only the folder in the first line. Alternatively, double-click START_ENGAGEIQ.cmd. Both entry points use the same setup script.

02 / What the launcher takes care of

Python: finds 64-bit Python 3.13. If missing, installs it for the current user through Windows winget.

Dependencies: creates .venv, installs the exact versions in requirements-lock.txt, and runs pip check. No activation command is required.

Data and models: checks bundled CSVs and trained artifacts. Missing raw data is downloaded from UCI; missing processed files or models are rebuilt. A model inference and analytics self-check run before startup.

Local site: reuses this workspace if already running, otherwise selects an available port between 5055 and 5065. The browser opens when the server is ready.

03 / Permissions, startup and stopping

Administrator access, firewall changes and live Facebook permissions are not required. The script needs write access to its own folder and internet for first-time installation or missing downloads. It binds only to 127.0.0.1.

ExecutionPolicy Bypass applies only to this launched PowerShell process. It does not permanently change your Windows policy. Organization-enforced restrictions are not overridden. Python installation uses the standard winget package and accepts its package/source terms.

Keep the terminal open. The printed URL is usually http://127.0.0.1:5055. Press Ctrl+C to stop the server. Saved uploads and prediction history remain in instance/.

Understand the workspace.

Real historical Facebook data, with analysis and prediction kept separate.

04 / What EngageIQ does

Workflow	What to explore
Historical analysis	Overview, format comparison, monthly trends, reaction mix, posting-time heatmap and highest-interaction posts. Content/date filters update the results. Export downloads the filtered posts.
Prediction	Enter page likes, post format, category, month, weekday, hour and paid status. The saved model estimates interactions; save scenario A and run a second estimate to compare.
Pattern discovery	K-Means groups posts by reactions, comments and shares. The interface shows group counts and profiles.
Your own CSV	Upload up to 8 MB / 100,000 rows. The app reports valid and removed rows. Uploads enable descriptive analysis and do not retrain the predictor.

05 / Data and model explanation

Analytics: the downloaded UCI Facebook Live Sellers file contains 7,050 posts, observed from July 2012 to June 2018. Total interactions = reactions + comments + shares. Likes are already part of reactions. No reach-based engagement rate is inferred.

Prediction: UCI Facebook Metrics contains 500 posts from one cosmetics brand in 2014. The model uses only seven pre-publication inputs. Weekday coding is Sunday = 1 through Saturday = 7. Analytics weekdays are calculated independently from timestamps.

Training uses months 1-9 (345 posts); months 10-12 (155 posts) remain held out for evaluation. Expanding-month validation compares Ridge, Decision Tree, Random Forest and Gradient Boosting with raw/log targets, plus a median baseline. Log-target Ridge was selected.

Held-out measure	Measured value
Mean absolute error (MAE)	135.86 interactions
Root mean squared error	252.03 interactions
R-squared	-0.011
Median baseline MAE	136.71 interactions

The model barely improves on the median baseline. Negative R-squared and high errors remain visible. These are historical research estimates, not guaranteed current performance. MAE is an average error measure, not a confidence interval. Scenario differences do not prove causal effects.

06 / Architecture

Browser (HTML, CSS, JavaScript, local Chart.js) -> Flask API -> Pandas analytics or saved scikit-learn pipeline. SQLite stores predictions. CSVs, models, charts and this PDF ship with the project. Cloud deployment is excluded.

Keep it running.

Useful commands, supported uploads and recovery guidance.

07 / Optional terminal commands

Install and verify everything without starting the website:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\run.ps1 -SetupOnly
```

Run the site without opening a browser automatically:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\run.ps1 -NoBrowser
```

Run the project tests after setup:

```
.\.venv\Scripts\python.exe -m pytest -q
```

08 / CSV format

```
post_id,post_type,post_date,likes,comments,shares
a,photo,2020-01-01 12:00,10,2,3
b,video,2020-01-02 18:00,20,4,6
```

Use photo, video, status or link for post_type. Supply reactions instead of likes if total reactions are available. Native UCI names are also accepted. Invalid rows and duplicates are removed and reported. Real viral outliers are retained. Example rows above illustrate the upload format only; they are not model training data.

09 / Troubleshooting

Message or symptom	Action
Python / winget missing	Install 64-bit Python 3.13 from python.org or install Windows App Installer for winget, then rerun.
Package download / certificate failure	Check internet access, system date and your proxy/certificate setup. Rerun to repair partial installation. The launcher preserves TLS verification.
Existing .venv uses another Python	Stop the site, rename .venv to .venv-old, then rerun. The launcher creates a compatible environment.
Port 5055 occupied	Use the URL printed by the launcher. It tries ports through 5065 and never stops another project.
Script blocked by organization policy	Ask your administrator to allow this local script. The launcher does not bypass managed restrictions.
Missing data or model	Rerun the launcher. Missing raw data needs internet; models can rebuild from the bundled datasets.

10 / Sources and files

Data: Dehouche (2018), Facebook Live Sellers in Thailand, DOI 10.24432/C5R60S; Moro, Rita & Vala (2016), Facebook Metrics, DOI 10.24432/C5QK55. Both are UCI datasets licensed CC BY 4.0. Full source links and checksums are included in docs/DATASET_SOURCES.md and data/raw/sources.json.

See README.md for development details and docs/API_DOCUMENTATION.md for endpoints. Keep instance/ private: it stores the browser session secret, uploads and SQLite history. The ZIP excludes private runtime files and the virtual environment.